Subnetting

1. Divide 192.168.10.0/24 into 4 Equal Subnets

Given network: 192.168.10.0/24

Step 1: Calculate the bits to borrow.

2^n=4 n=2

We borrow 2 bits from the host portion.

Step 2: Calculate the new subnet mask.

24+2=26

New subnet mask: 255.255.255.192

Step 3: Calculate the block size.

256-192=64

So, each subnet contains 64 IP addresses.

Subnet	Network Address	Subnet Mask	Valid Host Range	Broadcast Address
1	192.168.10.0/26	255.255.255.192	192.168.10.1–62	192.168.10.63
2	192.168.10.64/26	255.255.255.192	192.168.10.65–126	192.168.10.127
3	192.168.10.128/26	255.255.255.192	192.168.10.129–190	192.168.10.191
4	192.168.10.192/26	255.255.255.192	192.168.10.193–254	192.168.10.255

Final answer: 4 subnets, 64 total addresses per subnet and 62 usable host addresses per subnet.

2. Assign IP Addresses to Five Devices

We will use Subnet 1: 192.168.10.0/26.

- Network address: 192.168.10.0
- Broadcast address: 192.168.10.63
- Valid host range: 192.168.10.1–192.168.10.62
- Subnet mask: 255.255.255.192

Device	IP Address	Subnet Mask	Default Gateway
Router	192.168.10.1	255.255.255.192	Not required for this LAN interface
Laptop 1	192.168.10.10	255.255.255.192	192.168.10.1

Laptop 2	192.168.10.11	255.255.255.192	192.168.10.1
Printer	192.168.10.12	255.255.255.192	192.168.10.1
Mobile	192.168.10.13	255.255.255.192	192.168.10.1

All five addresses are valid and unique within this subnet.

3. Network Diagram

The diagram below shows Subnet 1, with the router connected to a switch and four end devices.

Router / Default Gateway

192.168.10.1/26

Network Switch

Laptop 1

192.168.10.10

Laptop 2

192.168.10.11

Printer

192.168.10.12

Mobile

192.168.10.13

Subnet: 192.168.10.0/26 · Usable range: 192.168.10.1–192.168.10.62

The other three subnets can be used for additional departments, device groups or separate LANs. A router or Layer 3 switch can connect them when routing is configured.

4. Step-by-Step Subnetting Calculation

Step 1: Understand the original network.

192.168.10.0/24 has 24 network bits and 8 host bits.

Step 2: Find the number of bits to borrow.

We need four subnets.

2^2=4

Borrow 2 bits from the host portion.

Step 3: Find the new CIDR prefix.

24+2=26

The new prefix is /26, and its subnet mask is 255.255.255.192.

Step 4: Calculate addresses per subnet.

The remaining host bits are:

32-26=6

Total addresses:

2^6=64

Usable host addresses:

64-2=62

We subtract 2 because the network and broadcast addresses cannot be assigned to ordinary hosts in this IPv4 subnet.

Step 5: Calculate the block size.

256-192=64

The network addresses increase by 64 in the last octet: 0, 64, 128, and 192.

Step 6: Find the broadcast addresses.

The broadcast address is one less than the next subnet's network address. Therefore, the broadcasts are .63, .127, .191, and .255.

Step 7: Assign device IP addresses.

Choose any five unique usable addresses from the same subnet. For Subnet 1, we assigned .1 to the router and .10 through .13 to the other devices.

Conclusion: The /24 network is divided into four /26 subnets, each containing 64 total addresses and 62 usable host addresses. All five devices are assigned valid addresses in Subnet 1.